

Assignment-01

1A) Problem :- Identify regions where people are at high risk of poverty

Input Data :- Income, employment, education, housing, population, healthcare access and living conditions

Algorithm :- Decision Tree

Working :-



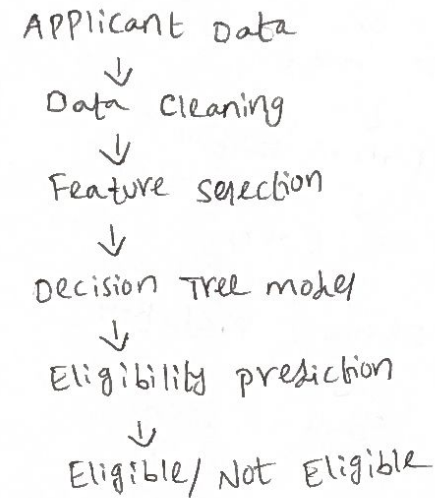
Advantages :- Easy to understand interpretable and useful for Policy Planning

Limitations :- Poor-quality or incomplete socio-economic data can reduce accuracy

2A) Problem :- Automatically identify people eligible for government welfare schemes

Features :- Income, family size, employment, age, disability status, housing and existing benefits.

Algorithm :- Decision Tree



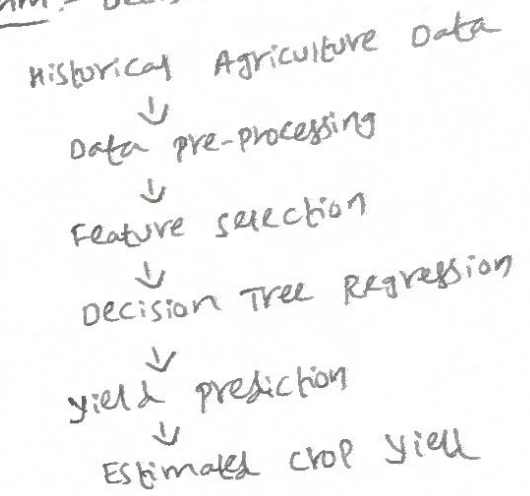
Advantages :- Reduces manual work and processing time

Limitations :- Bias in historical application data may produce under predictions

3A) Problem :- Predict crop yield using historical agriculture data

Features :- Rainfall, temperature, soil type, fertilizer, irrigation, crop type and previous yield.

Algorithm :- Decision Tree



Advantages :- Supports agricultural planning and food security decisions

Limitations :- unexpected weather conditions can affect predictions.

4A) Problem :- detect diseases from crop leaf images

Input :- Images of healthy and diseased leaves

Algorithm :- Decision Tree



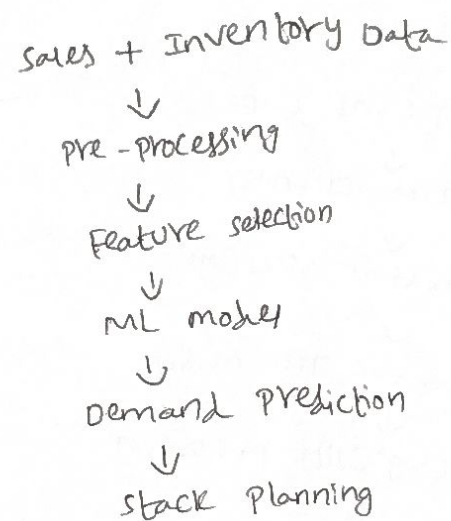
Advantages :- helps farmers take timely action

Limitations :- poor lighting and image quality may affect prediction

5A) Problem :- predict product demand to prevent over stocking and wastage

Feature :- sales history, season, price, holidays, product type and inventory.

Algorithm :- Decision Tree



Advantages :- Reduces food waste and improves inventory management

Limitations :- sudden changes in customer demand may reduce accuracy